

Smart Parking Management System*

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Abstract—We are solving the problem of manual and unsafe parking management by building a smart parking system that can control entry and exit without human assistance. The goal is to reduce delays, prevent unauthorized access, and improve overall safety in the parking area, including efficient vehicle movement between floors.

We implemented an automated gate control mechanism that locks and unlocks based on system decisions, removing the need for manual operation. In addition, the system includes a vehicle lift that automatically arrives when the ground floor is full and transfers vehicles to the upper level to increase parking capacity. The system also provides safety features such as fire detection, where a siren is triggered immediately and access points automatically unlock to support quick evacuation.

The system operates efficiently in real time, enabling fast gate and lift responses and reducing waiting time compared to manual control. In emergency situations like fire detection, it quickly activates alerts and unlocks access points, improving safety and response time.

Index Terms—IoT-based Smart Parking, Elevator-based Parking, Vehicle Lift Control, Vertical Parking, Automated Barrier Gate, Sensor Integration, Fire Alarm System, Real-Time Monitoring

I. INTRODUCTION

[Problem Statement] Many parking areas are still managed manually, which often causes long waiting lines at entry and exit points, increases the risk of human error, and weakens security by allowing unauthorized access. Another major issue occurs when the ground-floor parking becomes full: vehicles get congested because there is no automated way to transfer cars to higher floors. In addition, safety is a critical concern,

since in emergencies such as fire, delayed manual unlocking of gates and exits can slow down evacuation and increase risk. The motivation behind this project is to reduce time loss, improve parking security, and ensure a faster and safer response during emergencies. Therefore, we chose to implement an automated smart parking management system where the gate can lock and unlock without human assistance, and a vehicle lift can automatically arrive when a car approaches the lift area after the ground floor is full, transferring the vehicle to the upper level. Moreover, the system integrates fire safety by triggering a siren and automatically unlocking access points to support quick evacuation. This project was selected because it addresses a real-world problem and combines automation, capacity management, and safety features into a single efficient solution.

[Literature Review] Previous studies and prototype systems on smart parking mainly focus on automating parking management and reducing human effort. A common approach is to use sensors (e.g., ultrasonic or IR) to detect whether a parking slot is occupied and then provide real-time availability information through a display, web dashboard, or mobile application. This helps drivers locate empty spaces faster, reduces unnecessary searching inside the parking area, and improves overall traffic flow within the facility.

Several works also address automated access control at parking entrances and exits. In these systems, technologies such as RFID cards, QR-based authentication, or similar identification methods are used to open and close barrier gates automatically, minimizing unauthorized entry and decreasing waiting time caused by manual gate operation. Such access-control-based

parking solutions improve security and provide a smoother entry/exit experience.

Due to limited space in urban environments, some research and implementations propose multi-level or vertical parking concepts to increase parking capacity. These solutions often use mechanical stacking systems or elevator-based vehicle transfer mechanisms to move cars to upper levels, which can reduce congestion when the ground floor becomes full and allow more vehicles to be accommodated within the same footprint.

Safety and emergency response have also been considered in related works, particularly through fire or smoke detection and alarm/siren systems. While many implementations provide basic alert mechanisms, fewer systems integrate emergency actions directly into the parking control logic, such as automatically unlocking gates or access points to support rapid evacuation.

Based on these prior works, our project integrates multiple key features into a single system: automated gate locking/unlocking for secure entry and exit, an elevator-based vehicle lift that transfers cars to upper levels when the ground floor capacity is reached, and a fire safety mechanism that triggers a siren and automatically unlocks access points during emergencies. This combined approach aims to improve efficiency, security, capacity management, and emergency responsiveness in a practical smart parking environment.

[Innovative features added to your projects] Most existing smart parking ideas focus mainly on slot monitoring or basic gate automation. In many cases, the system only shows whether parking is full or not, but it does not actively manage vehicle movement across floors, and emergency safety actions are often not integrated into the same control logic.

Our project differs by combining end-to-end automation with capacity management and safety response in a single integrated system. When the ground floor becomes full, the system automatically dispatches a vehicle lift when a car approaches the lift area, transferring the vehicle to the upper level to reduce congestion and increase usable parking capacity. In addition, our system includes an emergency fire-response mechanism where a siren is activated and gates/access points are automatically unlocked to support fast evacuation. A real-time LED dashboard is also implemented to display the current number of parked vehicles, total slot capacity, and the number of available (vacant) slots, helping drivers and authorities make quick decisions.

Summary of Features:

- Automated gate locking/unlocking for entry and exit without human assistance.
- Vehicle lift integration for automatic transfer to upper floors when the ground floor is full.
- Real-time LED dashboard showing total slots, occupied slots (number of cars), and vacant slots.
- Fire/Gas safety mechanism with siren alert and automatic unlocking of gates/access points during emergencies.
- Integrated control workflow that improves efficiency, security, and emergency responsiveness.

II. PROPOSED METHOD

A. Block diagram of the overall system

III. PROPOSED METHOD

A. Block diagram of the overall system

Our proposed smart parking system is designed around an ESP32 microcontroller that collects data from multiple sensors and controls different actuators to automate gate operation, vehicle lifting, and emergency safety response. The overall system contains four major subsystems: (i) Entry/Exit gate control, (ii) Vehicle counting and LED dashboard display, (iii) Vehicle lift control for capacity management, and (iv) Fire safety and emergency unlocking.

Blocks to include in the overall block diagram (Power-Point):

- **ESP32 (Main Controller)**
- **RFID Reader** → (authentication for gate access)
- **IR Sensor (Entry)** and **IR Sensor (Exit)** → (vehicle counting)
- **Servo Motor (Gate)** ← (gate open/close control)
- **LED Display/Dashboard** ← (occupied/vacant slots information)
- **IR Sensor (Lift Front)** → (detect vehicle near lift)
- **Motor Driver** ← and **Lift Motor** ← (lift movement control)
- **Smoke Sensor** and **Flame Sensor** → (fire detection)
- **Buzzer/Siren** ← (emergency alarm)
- **Emergency Gate/Exit Unlock (2nd Floor)** ← (emergency evacuation support)

How the blocks are connected (use arrows in the diagram):

- RFID Reader, Entry IR, Exit IR, Lift-front IR, Smoke Sensor, and Flame Sensor **send signals to ESP32**.
- ESP32 **outputs control signals** to the Gate Servo Motor, Motor Driver (Lift Motor), LED Display, Buzzer/Siren, and Emergency Unlock mechanism.

B. Detailed explanation of the block diagram

(1) ESP32 Controller: The ESP32 acts as the core processing unit of the system. It continuously reads the RFID module and all sensor inputs (IR sensors, smoke sensor, flame sensor) and executes the decision logic. Based on these inputs, the ESP32 generates control outputs to operate the gate servo, lift motor driver, LED dashboard, and emergency alarm/unlocking mechanisms.

(2) RFID-based Automated Gate Control (Servo Motor): At the parking entry point, the RFID reader is used for authentication. When a valid RFID tag/card is detected, the ESP32 triggers the servo motor to unlock/open the gate. After the vehicle passes, the gate is closed again by commanding the servo motor back to the locked position. This ensures entry/exit operation without human assistance and improves access security.

(3) Vehicle Counting using Entry and Exit IR Sensors: Two IR sensors are installed for counting vehicles: one at the entry lane and another at the exit lane. Whenever a vehicle is

detected at the entry IR sensor, the ESP32 increases the parked vehicle counter; similarly, when a vehicle is detected at the exit IR sensor, the counter is decreased. Using this counter and a predefined total slot capacity, the system can determine the number of occupied and vacant slots in real time.

(4) LED Dashboard Display: The LED display acts as a real-time dashboard. The ESP32 updates the display with key information such as total slots, current number of parked vehicles (occupied slots), and available/vacant slots. If the ground floor reaches full capacity, the display can show a *FULL* or similar status message to inform users.

(5) Lift Control for Full-Capacity Situation: When the ground floor becomes full, the system uses a dedicated IR sensor placed in front of the lift to detect the presence of a vehicle approaching the lift area. If the lift-front IR sensor detects a vehicle (and the system indicates ground floor is full), the ESP32 sends control signals to a motor driver, which drives the lift motor to move the vehicle upward to the next floor. This mechanism reduces congestion and helps manage parking capacity automatically across floors.

(6) Fire Safety and Emergency Response (Smoke + Flame Sensors): For safety, both smoke and flame sensors are used to detect possible fire incidents. If any of these sensors indicates a fire condition, the ESP32 immediately activates the buzzer/siren to alert users. At the same time, the system switches to an emergency mode where all gates are unlocked/opened automatically to support rapid evacuation. Additionally, an emergency exit/unlock arrangement is provided for the upper floor (2nd floor) so that people can safely move down and exit during an emergency.

Overall, the proposed method integrates access control, real-time monitoring, vertical vehicle transfer, and emergency safety into one unified ESP32-based automated system.

IV. IMPLEMENTED HARDWARE SYSTEM

A. List of the hardware and software Environment

Hardware Environment: The implemented smart parking system is built using an ESP32 microcontroller as the main control unit. The system uses an RC522 RFID reader for authentication-based gate access. IR sensor modules are used at the entry and exit points for vehicle counting, and an additional IR sensor is placed in front of the lift to detect vehicle presence. A servo motor is used to open/close the parking gate, while a motor driver (L298N) controls the lift motor for vertical vehicle transfer. For safety, an MQ-2 smoke sensor and a flame sensor module are integrated with a buzzer/siren to generate emergency alerts and trigger automatic unlocking. An LED display is used as a dashboard to show the total capacity, occupied vehicles, and available (vacant) parking slots.

Software Environment: The firmware was developed using Arduino IDE and uploaded to the ESP32. The system logic was implemented in C/C++ (Arduino framework), using standard libraries for RFID communication, display handling, and GPIO-based sensor/actuator control.

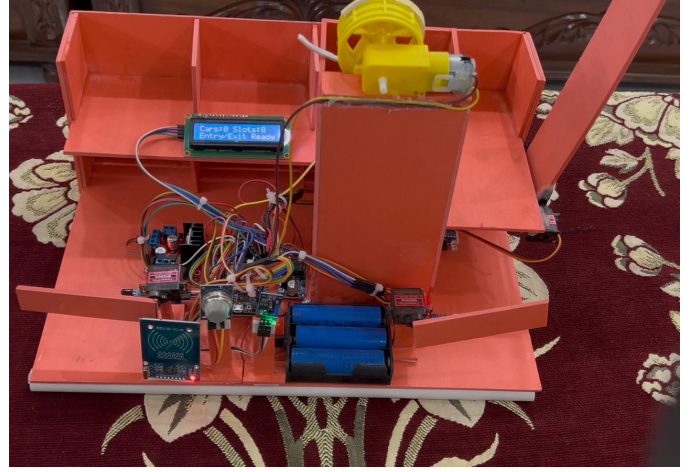


Fig. 1. Real photograph of the implemented smart parking hardware prototype (annotated to indicate the main modules).

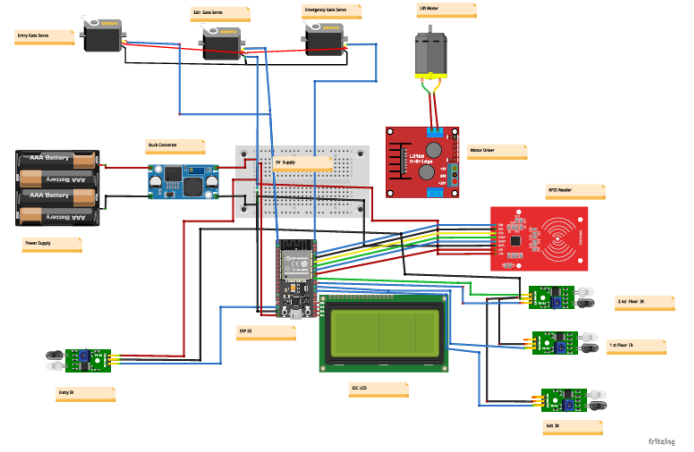


Fig. 2. Wiring and interconnection diagram of the implemented system (ESP32, RFID, IR sensors, gate servos, lift driver/motor, power, and display connections).

B. Figures of the implemented hardware system

This subsection presents the implemented hardware system using two figures: (i) a real photograph of the complete prototype setup, and (ii) the wiring/connection diagram prepared for the system. The figures are annotated (labeled) so that the main modules can be identified easily, including the ESP32 controller, RC522 RFID reader, IR sensors (entry/exit/lift-front), gate servo motors, lift motor driver and motor, power supply, and the display unit.

Fig. 1 shows the real implemented prototype arrangement, where the ESP32 is used as the central controller. The gate operation is performed using servo motors, and the RC522 RFID reader is used for authentication-based access. IR sensors are placed at the entry and exit paths for vehicle counting and at the lift-front area for lift triggering. The display unit shows the real-time occupancy information (total slots, occupied cars, and vacant slots), and the alarm modules are used for safety alerts.

Fig. 2 illustrates the complete wiring and module-level connectivity of the system. All sensors provide input signals to the ESP32, and the ESP32 outputs control signals to the servo motors (gate), motor driver (lift motor), and display. The diagram helps readers understand how each subsystem is electrically connected and how the control flow is implemented through the ESP32-based architecture.

Brief Figure Description: Overall, the figures demonstrate an integrated hardware implementation where the ESP32 continuously reads sensor inputs, authenticates vehicles through RFID, updates parking occupancy and available slots on the display using entry/exit IR-based counting, controls gate opening/closing via servos, operates the lift through a motor driver when required, and supports safety response through alarm-based emergency actions.

C. Figures and Tables

If you want to add figures or tables, use the syntax of the codes from this section!

a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. ??”, even at the beginning of a sentence.

The implemented smart parking management system was tested under multiple operating scenarios to verify the correct functioning of each subsystem: (i) RFID-based gate access, (ii) IR-sensor-based vehicle counting and LED dashboard update, (iii) lift triggering when the ground floor is full, and (iv) emergency safety response using smoke and flame sensors. The system outputs were observed from the physical operation of the actuators (servo gate, lift motor), the LED display readings, and the alarm behavior.

D. RFID-based Gate Access Result

The RC522 RFID reader successfully identified authorized tags/cards and the ESP32 executed the access logic accordingly. For a valid RFID tag, the gate servo performed the unlocking/opening action and then returned to the locked/closed position after the vehicle passed. For an invalid or unauthorized tag, the system denied access and the gate remained locked, demonstrating basic security enforcement through authentication.

E. Vehicle Counting and LED Dashboard Result

The entry and exit IR sensors were used to count vehicles entering and leaving the parking area. When a vehicle crossed the entry sensor, the occupied vehicle count increased; when a vehicle crossed the exit sensor, the count decreased. The LED dashboard updated these values in real time and continuously displayed the current occupancy information, including total slot capacity, number of parked vehicles (occupied slots), and available (vacant) slots. This real-time update reduced ambiguity for users and made the parking status visible at a glance.

TABLE I
HARDWARE COMPONENTS USED IN THE IMPLEMENTED SYSTEM

Hardware Component	Qty	Purpose / Role
ESP32 Microcontroller	1	Main controller that reads sensors and controls gate, lift, display, and emergency actions
RFID Reader (RC522)	1	Authenticates authorized access and triggers gate opening
IR Sensor (Entry)	1	Detects vehicles at the entry point for counting
IR Sensor (Exit)	1	Detects vehicles at the exit point for counting
IR Sensor (Lift Front)	1	Detects vehicle presence near the lift area to trigger lift operation
Servo Motor (Gate)	1	Opens/closes (locks/unlocks) the parking gate automatically
Motor Driver (L298N)	1	Controls the lift motor based on ESP32 control signals
Lift Motor (DC Motor)	1	Moves the lift to transfer vehicles to the upper floor
Smoke Sensor (MQ-2)	1	Detects smoke/gas to indicate an emergency fire condition
Flame Sensor Module	1	Detects flame/fire to indicate an emergency condition
Buzzer/Siren	1	Provides audible alarm during emergency situations
LED Display (Dashboard)	1	Displays total slots, occupied cars, and vacant slots in real time
Emergency Unlock Mechanism (2nd Floor)	1	Supports emergency exit/unlock during fire or hazard
Power Supply Unit	1	Supplies required power to controller, sensors, and motors

F. Full-Capacity Handling with Lift Result

When the occupied vehicle count reached the predefined ground-floor slot capacity, the system indicated that the ground floor was full (vacant slots became zero). In this situation, if a vehicle approached the lift area, the lift-front IR sensor detected the vehicle and the ESP32 triggered the motor driver (L298N) to operate the lift motor. The lift mechanism then transferred the vehicle to the upper floor, reducing congestion and enabling efficient capacity management across floors.

G. Safety and Emergency Response Result

For safety testing, both the MQ-2 smoke sensor and the flame sensor module were used to detect hazardous conditions. Under normal conditions, the buzzer/siren remained OFF. When smoke or flame was detected beyond the configured threshold, the ESP32 immediately activated the buzzer/siren and switched the system into an emergency mode. In this

mode, access points were automatically unlocked to support quick evacuation, including the emergency arrangement for the upper floor. This behavior confirms that the system provides an automated fail-safe response during emergency situations. Overall, the experimental observations confirm that the integrated system can automate parking access, maintain real-time occupancy information, manage full-capacity situations using a lift mechanism, and respond promptly to emergency conditions.

V. DISCUSSION

The implemented system demonstrates a practical low-cost approach to automated parking management with integrated safety features. However, like any prototype, it has limitations related to sensor reliability, environmental noise, and mechanical constraints of the lift subsystem. This section discusses these limitations, trade-offs, alternative designs, and the broader impact of the solution.

A. Limitations and Cases Where the System May Not Work as Expected

The performance of IR sensors can be affected by strong ambient light, reflective vehicle surfaces, or misalignment, potentially causing missed detections or false triggers. Such counting errors can lead to incorrect occupied/vacant slot values on the LED dashboard. RFID authentication reliability may also vary depending on tag orientation, interference, or reading distance. In addition, the lift subsystem depends on mechanical stability, motor torque, and proper calibration; overloading, vibration, or misalignment can reduce reliability. For safety sensing, the MQ-2 and flame sensors can produce false alarms in environments with dust, smoke-like fumes, or external heat sources, so careful placement and threshold tuning are important.

B. Conflicting Requirements and Trade-offs (P2)

A key trade-off in this project was balancing cost, complexity, and reliability. Low-cost modules (IR sensors, RFID, and simple motor control) make the system affordable and easier to build, but more advanced sensing and control solutions could improve accuracy and robustness at a higher cost and with increased implementation complexity.

C. Alternative Solutions / Designs (P3)

Two alternative designs could be considered in future improvements. First, camera-based monitoring and license plate recognition could replace RFID to reduce dependence on tags/cards; however, it would require higher computational resources, better lighting conditions, and significantly higher cost. Second, per-slot occupancy sensing (e.g., ultrasonic sensor per slot or load sensors) could provide more accurate slot availability information compared to entry/exit counting, but it would increase wiring complexity, maintenance effort, and overall hardware cost.

D. Impact of the Solution (P4)

The proposed system can improve parking efficiency by reducing manual gate operation and providing real-time occupancy information, while the lift mechanism increases capacity utilization in space-constrained environments. The integrated emergency response improves safety by enabling quick alarms and automatic unlocking during hazardous situations. This solution is applicable to real-world settings such as apartment buildings, commercial parking areas, and campus parking facilities.

E. Applicable Standards, Manuals, and Ethical Considerations (P5)

During implementation, module manuals and interfacing guidelines were followed to connect and configure devices such as the RC522 RFID reader, MQ-2 smoke sensor, and motor driver safely. Ethical considerations include minimizing unnecessary collection of personal data; the system can be designed to verify RFID access without storing sensitive user identity information beyond what is needed for access control.

F. Stakeholders and Requirement Incorporation (P6)

Potential stakeholders for this system include drivers, building/security management, and facility operators. Their common requirements are fast entry/exit, clear visibility of available slots, reduced congestion when the ground floor is full, and reliable safety response. These requirements were incorporated through automated gate control, a real-time LED dashboard, lift-based capacity management, and emergency unlocking features.

G. Interdependence and Modularity of System Components (P7)

The system is organized into modular subsystems: gate access control (RFID + servo), counting and display (entry/exit IR + LED), lift control (lift-front IR + motor driver), and safety response (MQ-2/flame + buzzer + emergency unlock). While each subsystem has its own sensors and outputs, they are coordinated through the ESP32 controller, allowing improvements or replacement of one module without redesigning the entire system.

In summary, the system provides an effective integrated solution for automated parking management and safety, with clear opportunities for future enhancement in sensing accuracy, mechanical reliability, and advanced authentication or monitoring features.]

VI. RESULTS

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In summary, the system provides an effective integrated solution for automated parking management and safety, with clear opportunities for future enhancement in sensing accuracy, mechanical reliability, and advanced authentication or monitoring features.

VIII. CONCLUSION

This project presents an ESP32-based smart parking management system that automates parking entry/exit using RFID authentication and servo-controlled gate operation while maintaining real-time vehicle counting through IR sensors. The system also supports vertical parking by triggering a lift mechanism when the ground floor reaches full capacity, and it enhances safety by activating an alarm and automatically unlocking access points when smoke or flame is detected.

The system operates efficiently in real time by providing fast gate response, continuous LED dashboard updates for occupied/vacant slots, and quick emergency activation to improve overall safety and reduce congestion. As future work, the system can be improved by adding more robust sensing (e.g., per-slot detection or sensor fusion) to reduce false triggers and increase counting accuracy. Additional features such as Wi-Fi/app-based monitoring, event logging, and stronger mechanical lift control (e.g., limit switches and load sensing) can further enhance reliability and usability for real-world deployment.

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